Listing 11.6 Time travel reasoning/planning problem

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In a sci-fi film, Alex is a time traveler who decides to go back in time
to witness a famous historical battle that took place 100 years ago,
which lasted for 10 days. He arrives three days before the battle starts.
However, after spending six days in the past, he jumps forward in time
by 50 years and stays there for 20 days. Then, he travels back to
witness the end of the battle. How many days does Alex spend in the past
before he sees the end of the battle?
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While it may be somewhat disappointing to see the model get the wrong answer, it does, however, do a far better job of breaking down the problem and demonstrating its answer. Listing 11.7 shows the sample output from posing the problem in listing 11.6 to the Strawberry model. Note, you may get a different answer because of the stochastic nature of the LLM.

Listing 11.7 o1-preview response to time travel problem

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To determine how many days Alex spends in the past before witnessing the
end of the battle, let's reconstruct the timeline step by step:
Arrival in the Past:
Alex arrives 3 days before the battle begins.
He spends these 3 days waiting for the battle to start.
During the Battle:
The battle starts, and Alex experiences the first 3 days of it.
At this point, he has spent a total of 6 days in the past (3 days before
the battle and 3 days during the battle).
Time Jump Forward:     #1
After these 6 days, Alex jumps 50 years forward in time.
He spends 20 days in this future time.
Return to the Past:
Alex then returns to the past to witness the end of the battle.
Since the battle lasts 10 days and he left after day 3 of the battle,
there are 7 days remaining in the battle.
However, because he was away for 20 days, which is longer than the
remaining 7 days of the battle, he must return to a point in time before
the battle ends to witness its conclusion.
To see the end of the battle, he returns to the past on the last day of
the battle.
Final Days Spent Before the Battle Ends:
Upon returning, he spends 1 more day in the past to witness the battle's
end.
Total Days Spent in the Past Before Seeing the End:
6 days before jumping forward in time.
1 day after returning to the past.
Total: 6 + 1 = 7 days.
Answer: 7
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#1 It becomes obvious where the model is making the error.

Because we know the right answer is 27, we know the LLM is wrong, but if we didn’t, we could just as easily assume that the work and reasoning were all correct. Problems like this can happen when we remove feedback in